

When a Trading Edge Disappears

Causal Clocks, Selection Bias, and a Bitcoin Prediction Market Study

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Abstract

I investigated whether cross-venue Bitcoin microstructure could improve the probability quoted by a five-minute prediction market. The project recorded multiple exchanges, Polymarket order books, and oracle-related feeds; built 5 Hz episodes; trained a residual temporal convolutional network (TCN); simulated a two-slot strategy; and operated a guarded shadow system. Early results appeared profitable. Adversarial evaluation changed the conclusion.

The lowest observed test Brier score after inspecting seven calibration maps is 0.127732, versus 0.128276 for the market. The paired 388-market 95% interval for model minus market is $[-0.002730, 0.001721]$. A corrected White's Reality Check appends the combined reported winner to the original search matrix and gives $p=0.5667$ across 847 candidates. The historical episode builder also replayed messages at venue source time rather than local receive time, allowing late messages to appear before they were available. Therefore the positive backtest is a zero-latency counterfactual, not live-achievable evidence.

The final verdict is **NO DEMONSTRATED TRADING EDGE**. The contribution is a reproducible negative result and a concrete account of how causal clocks, dependence, selection, and execution assumptions can erase an apparent edge.

I conceived and directed the study, including the hypotheses, causal rules, audit design, failure criteria, and interpretation. AI tools supported much of the software implementation.

Research question

Can high-frequency cross-venue Bitcoin information improve the prediction market's own probability near resolution, and can any improvement survive realistic statistical and execution constraints?

Hypothesis	Operational test	Result
H1: TCN residual improves forecast	Five seeds, seven maps, paired cluster bootstrap	Not demonstrated
H2: rule survives search	847-candidate White's Reality Check	Fail, $p=0.5667$
H3: profit is live-achievable	Availability-clock and execution audits	Not demonstrated

1. System and data

The software includes asynchronous recorders for Polymarket and several centralized venues; manifest and quality layers; canonical data builders; model training; replay; shadow decisions; and statistical audits. The locally measured private inventory reports 246.7 billion compressed bytes, 119,665 files, and 76 recorded days between April 19 and August 9, with 37 missing calendar days. This is a committed inventory claim, not a publicly recountable corpus total.

The v2 modelling dataset contains 7,470 training, 917 validation, and 516 test markets. Each market is represented by 1,500 time steps at 200 ms cadence, 69 numerical features, and a validity-mask channel. Training ends June 16; validation covers June 23-27; test covers June 28-July 2. Full tensors remain private, but their hashes, feature metadata, normalization, and split contract are public.

The model is a seven-block causal residual TCN with 64 channels and a 101.8-second receptive field. It predicts a residual over `logit(p_market)`, clamped to ± 0.75 . Five fixed seeds share the same events. The public release includes all five checkpoints, all validation/test loss-band predictions, and a ten-market inference fixture.

2. The central methodological failure: availability time

An event has at least two relevant times: when the venue says it occurred and when the recorder received it. A live policy cannot act on the message before receipt. Historical state must therefore advance under availability time, while source time remains useful for staleness and latency diagnostics.

One late message, two historical realities

Source-time replay (historical build)



Receive-time replay (causal requirement)



Figure 1. Source time can reveal a late message before local availability.

The episode builder historically inserted Polymarket and centralized-exchange messages under venue source time. Receive timestamps were audit fields rather than the admission clock. If a source timestamp is 12:00 but recorder receipt is 12:02, the replay can expose the message two minutes early. A freshness check after admission cannot repair this.

The correct classification is **SOURCE-TIME COUNTERFACTUAL**. Model scores and strategy PnL can still describe a zero-latency counterfactual, but cannot establish information or profit available to a live participant. A valid correction requires raw receive-time reconstruction and retraining; that experiment has not been performed.

3. Forecast evaluation

Scoring design

The evaluation uses every valid 200 ms row with 15 to 150 seconds remaining: 66,050 validation rows and 96,494 test rows. Brier loss is paired against the market on the same row. Seven mappings were examined: identity; validation temperature, Platt L2, isotonic, and residual shrinkage; and train day block temperature and residual shrinkage.

For uncertainty, I average the five seed losses within each row, subtract the market loss, and resample whole markets with replacement while retaining cluster size. The primary 20,000-replicate interval has 388 test-market clusters. A five-day cluster bootstrap is only a sensitivity check because five clusters are too few for stable inference.

Test Brier difference vs market (x10,000; lower is better)

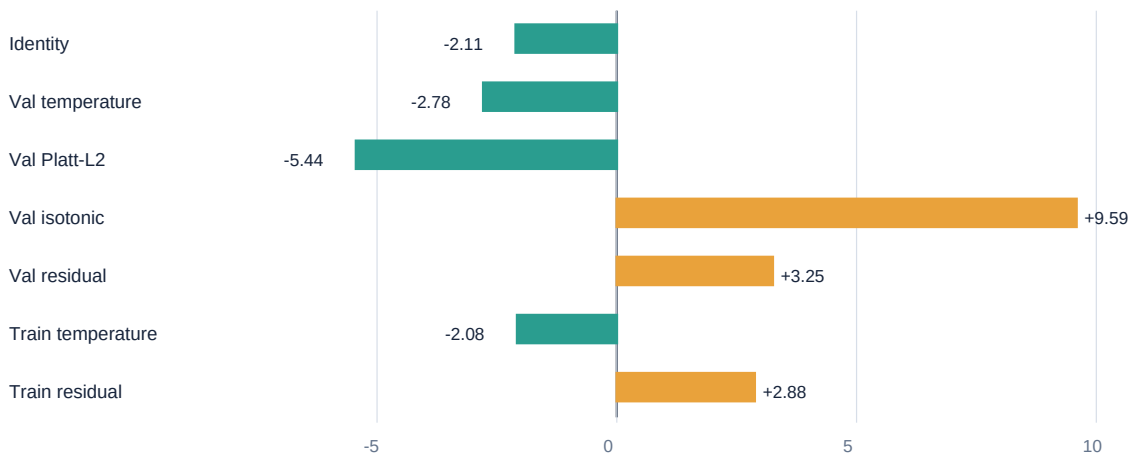


Figure 2. Point estimates vary across seven inspected calibration maps; the primary Platt-L2 cluster interval still crosses zero.

Test quantity	Value
Lowest observed model Brier (Platt L2)	0.127732
Market Brier	0.128276
Difference	-0.000544
Market-clustered 95% interval	[-0.002730, 0.001721]
Seeds individually below market	4 of 5

The interval includes zero. Isotonic calibration wins on validation but loses to the market on test. Selecting Platt L2 after inspecting all seven test rows is itself selection. The canonical verdict is **EXPLORATORY; IMPROVEMENT NOT DEMONSTRATED**. A confirmatory claim would need one frozen mapping, unseen receive-time data, and many more independent days.

4. Strategy search and the disappearing profit

The stored combined backtest reports 706 trades, +214.1461, and daily Sharpe 0.4085 over 32 available days. These values are internally consistent with the retained lock, but the strategy was chosen after extensive search.

The original anti-snooping matrix contains 846 candidate daily PnL series. Its stored report says $p=0.8355$, a digit that a fresh current-method run does not reproduce. More importantly, the matrix omits the combined strategy that became the reported winner. The public correction appends that series, recenters all 847 candidates, and draws 20,000 stationary bootstrap day paths with mean block length three.

Selection-adjusted null distribution of the best strategy

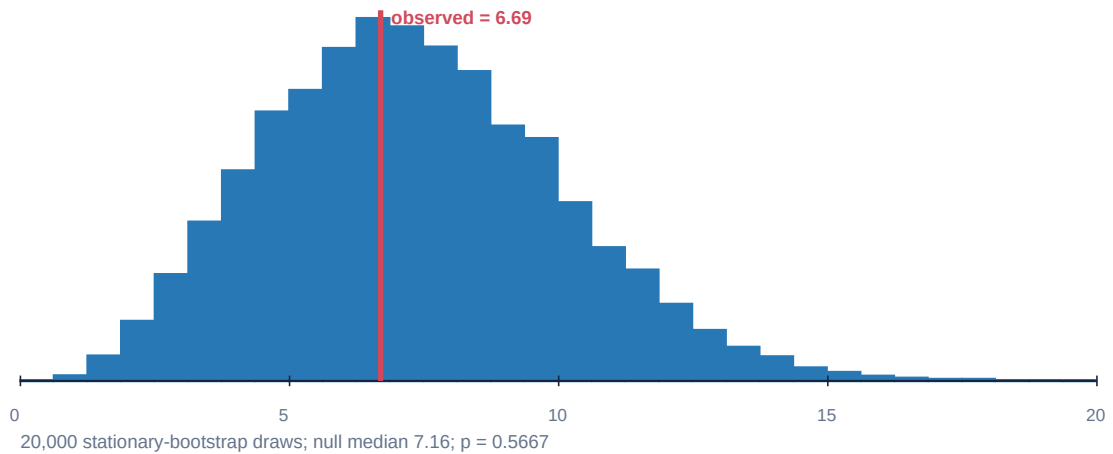


Figure 3. The observed winner is not extreme relative to the best result expected across 847 searched candidates.

Corrected audit quantity	Value
Candidates	847
Days	32
Observed best mean daily PnL	6.6921
White's Reality Check p-value	0.5667
Reject at 10%?	No

The individual locked early and late rules rank fifth and sixteenth in the old matrix; a different all-day early rule ranks second. The reported combined winner is not unusual relative to the best outcome expected from the searched family. Its verdict is **INVALID AS EDGE EVIDENCE**, even before the causal-clock problem is considered.

5. Execution and prospective operation

The historical execution audit does not rescue the strategy. It chooses books under source time, drops 20 missing books, includes 72 of 162 observations older than ten seconds, and reaches a maximum age of 62.497 seconds. Its adverse-fill stress changes the trade count from 706 to 722 instead of holding decisions fixed. Queue position, self-impact, and quote response are not modeled. The classification is **UNRESOLVED / INVALID AUDIT**.

A separate five-day post-lock artifact reports 262 offline-replay trades and +23.90, with a 90% day-bootstrap interval of [-15.95, 60.36]. The interval includes zero and the same source-time episodes determine its entries. It is an **INCONCLUSIVE SOURCE-TIME OFFLINE POST-LOCK REPLAY**, not a live forward score.

The strongest prospective evidence is operational rather than financial. Hashed local shadow logs from July 12-13 contain 45,436 decisions: 164 entries, 26,691 no-edge skips, and 18,581 source-health skips. Sanitized entry rows and complete denominators are public. They show that the guarded system ran and often refused to act. Local timestamps are not independent timestamps, and no fill or PnL conclusion follows.

6. Reproducibility and provenance

The public release is deliberately compact. From a fresh clone, a reader can recompute all seven Brier rows and cluster intervals, rerun the corrected 847-candidate selection audit, load all five checkpoints on real-shaped inputs, and verify that the claim register and PDF match their numerical sources.

Evidence level	Public material	Supported conclusion
Direct	Code, tests, predictions, checkpoints	Score and mini-inference consistency
Direct	847 candidate-day returns and null draws	Negative selection verdict
Bound	Split and private-array SHA-256	Named-input traceability only
Local	34 commit records and hashed July logs	Iterative development and operation
Missing	Raw corpus and receive-time retraining	No causal performance conclusion

The private-source history has 34 local commits from May 20 through June 29. It records positive claims, reversals after audit, and repeated bot-safety fixes. The July/August TCN and final audit phase was not committed there. Accordingly, chronology is **LOCAL PROVENANCE; NOT INDEPENDENTLY TIMESTAMPED**, not externally authenticated registration.

7. Conclusion

The apparent strategy profit disappears as evidence in three stages. Dependence-aware forecast uncertainty includes zero. The selected winner is ordinary relative to 847 searched candidates. The source-time replay cannot establish what was knowable live, and the execution audit is invalid.

The final verdict is **NO DEMONSTRATED TRADING EDGE**. This does not prove that prediction markets are perfectly efficient or that no causal signal could exist. It shows that this experiment did not establish one. The next experiment must use receive-time state, externally freeze one policy before new data, retain complete decision logs, and evaluate decision-matched execution.

The result I would carry forward is methodological: a smaller honest claim, with public failure analysis and reproducible uncertainty, is stronger research than a profitable number whose clock and selection process cannot support it.

References and artifact index

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- Politis, D. and Romano, J. (1994), *The Stationary Bootstrap*.
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- Selection evidence: selection_audit.json.
- Reproduction evidence: evaluation_repro_v2/manifest.json.
- Canonical evidence: publication_claims.json.